Lecture Material #7

Information Security

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Faculty Profile

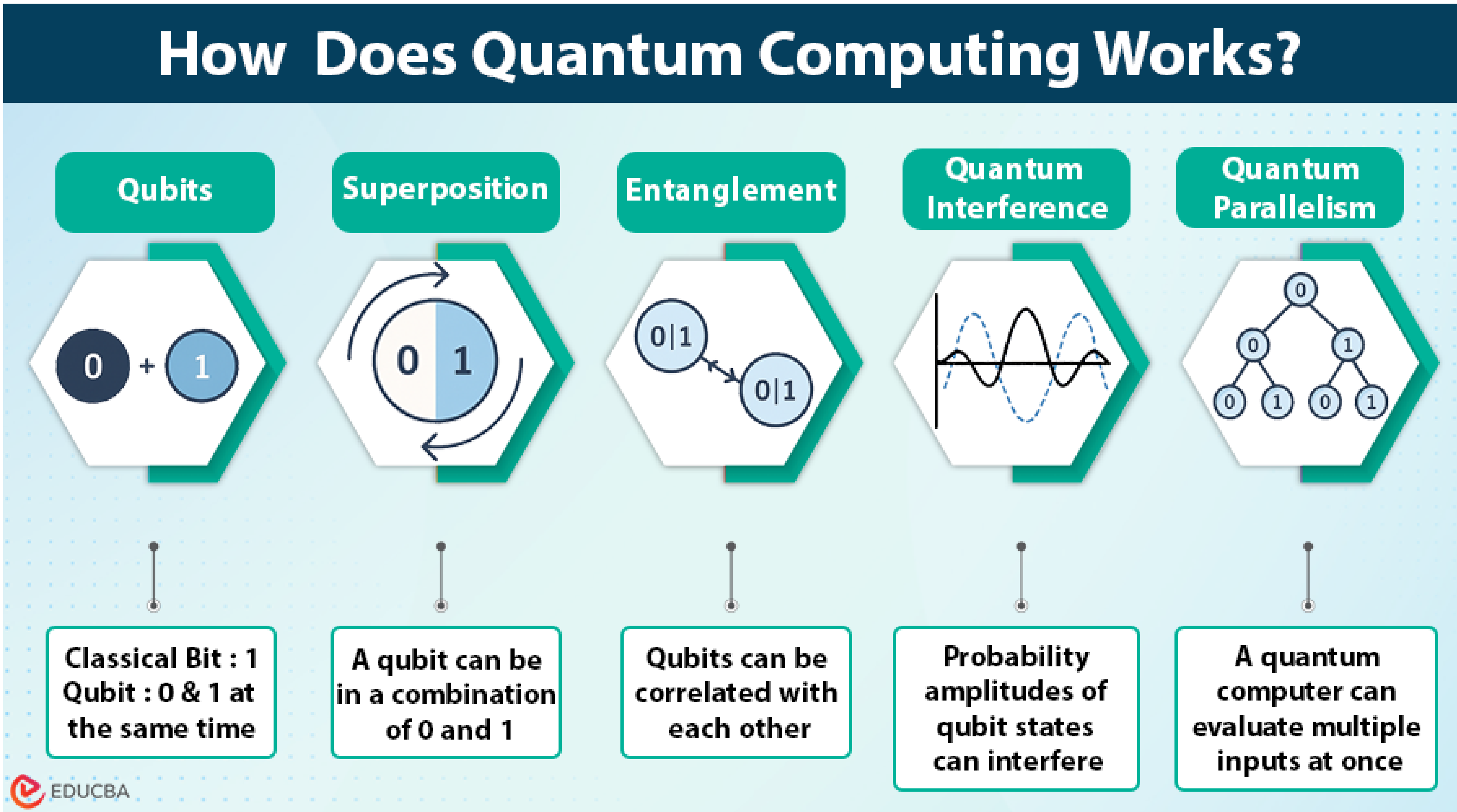
Quantum Attack

Quantum Computing

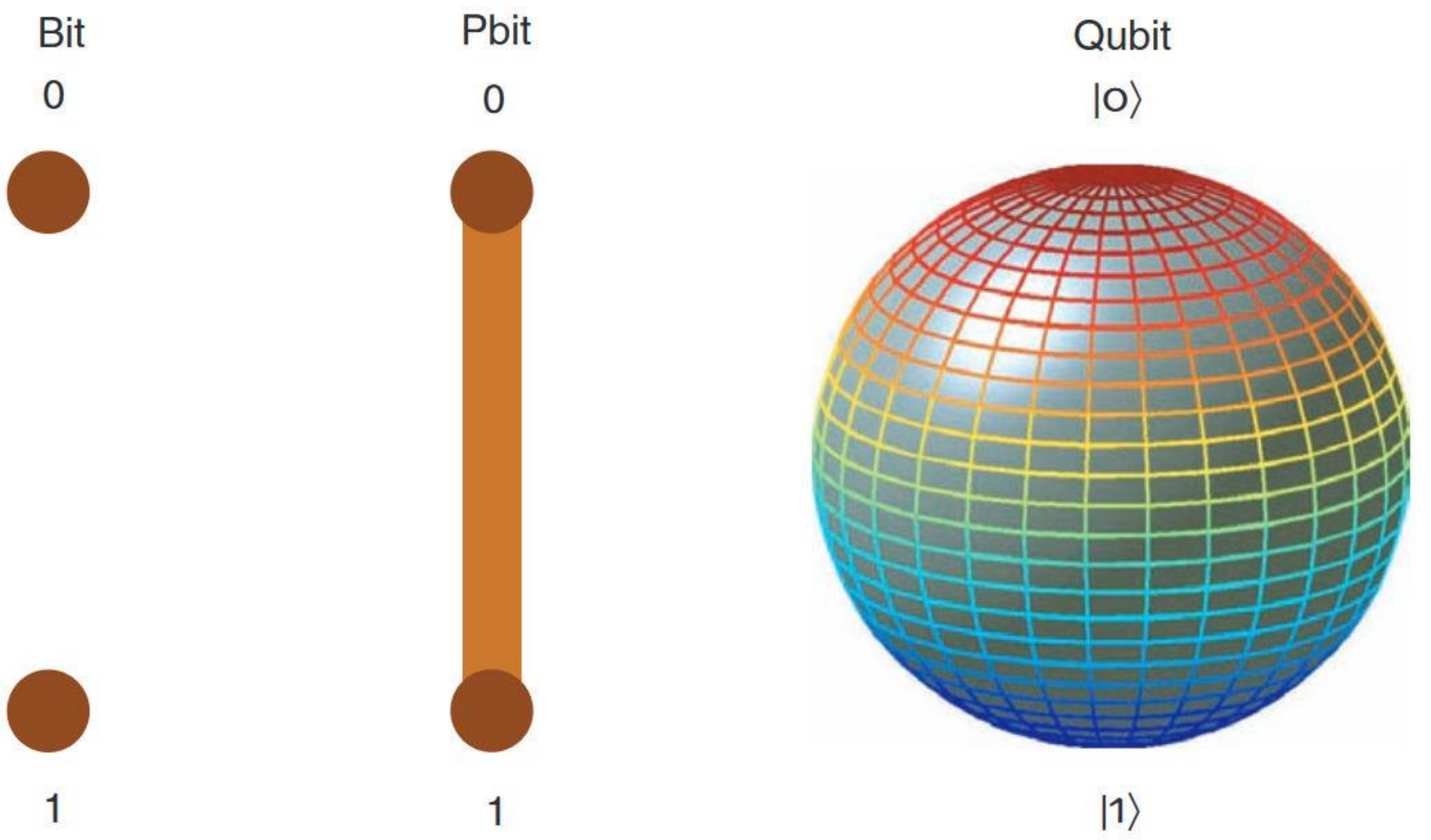
Key Principles of Quantum Computing

- **Qubits (Quantum Bits):** Unlike traditional binary bits (0 or 1), a qubit can be 0, 1, or a combination of both simultaneously, allowing for massive parallel processing.
- **Superposition:** Qubits can exist in a "spinning" state, acting as both 0 and 1 at the same time until measured.
- **Entanglement:** Qubits can be linked, meaning the state of one qubit instantaneously depends on another, even over distances, allowing for exponential increases in computational power.
- **Interference:** Quantum algorithms use interference to cancel out incorrect paths and amplify the correct answer.

Quantum Computing



Quantum Computing



Quantum Computing

Quantum Attack

Why Qubits Are Powerful

Because of superposition + entanglement:

- 1 qubit → represents 2 states
- 2 qubits → 4 states
- n qubits → 2^n states

This allows quantum computers to:

- Solve certain problems much faster than classical computers
- Work efficiently on cryptography, optimization, simulations, etc.

Quantum Computing

How It Differs from Classical Computing

- **Processing Speed:** While a classical computer checks possibilities one by one, a quantum computer can evaluate many possibilities simultaneously.
- **Scaling:** Every added qubit exponentially increases processing capacity, unlike the linear growth of traditional transistors.
- **Measurement:** When measured, the superposition collapses into a definite 0 or 1, providing the final result.

Quantum Attack

A **quantum attack** is a cryptographic attack that uses **quantum computers** and quantum algorithms to break or weaken classical cryptographic systems.

Why It Matters

- Quantum computers can solve certain mathematical problems **much faster** than classical computers.
- Many widely used cryptographic schemes rely on problems that are **hard for classical computers but easy for quantum ones.**

Key Quantum Algorithms

- **Shor's Algorithm** → breaks public-key cryptography
- **Grover's Algorithm** → weakens symmetric cryptography

Quantum Attack

Classical vs Quantum Computers

Classical Computers

- Limited by exponential or sub-exponential complexity
- RSA/ECC considered secure under classical assumptions

Quantum Computers

- Exploit **quantum parallelism and interference**
- Can solve factoring and discrete logarithms in **polynomial time**

Quantum Attack

Why RSA is Vulnerable to Quantum Attacks

RSA Security Depends On

- Difficulty of **integer factorization**
- Example: factoring large numbers (2048-bit keys)

Quantum Impact

- **Shor's Algorithm** can factor large integers efficiently
- RSA becomes **completely breakable** once large-scale quantum computers exist

Quantum Attack

Why ECC is Vulnerable to Quantum Attacks

ECC Security Depends On

- Elliptic Curve Discrete Logarithm Problem (ECDLP)

Quantum Impact

- Shor's Algorithm also solves discrete logarithms efficiently
- ECC loses its security advantage despite smaller key sizes

Quantum Attack

Why Symmetric Algorithms Are Less Vulnerable

Examples

- AES
- DES (already weak, not quantum-related)
- Hash functions (SHA-256, SHA-3)

Quantum Impact

- **Grover's Algorithm** provides only a **quadratic speedup**
- Security loss is manageable by increasing key size

Example

- AES-128 → equivalent to ~64-bit security
- AES-256 → still considered quantum-resistant

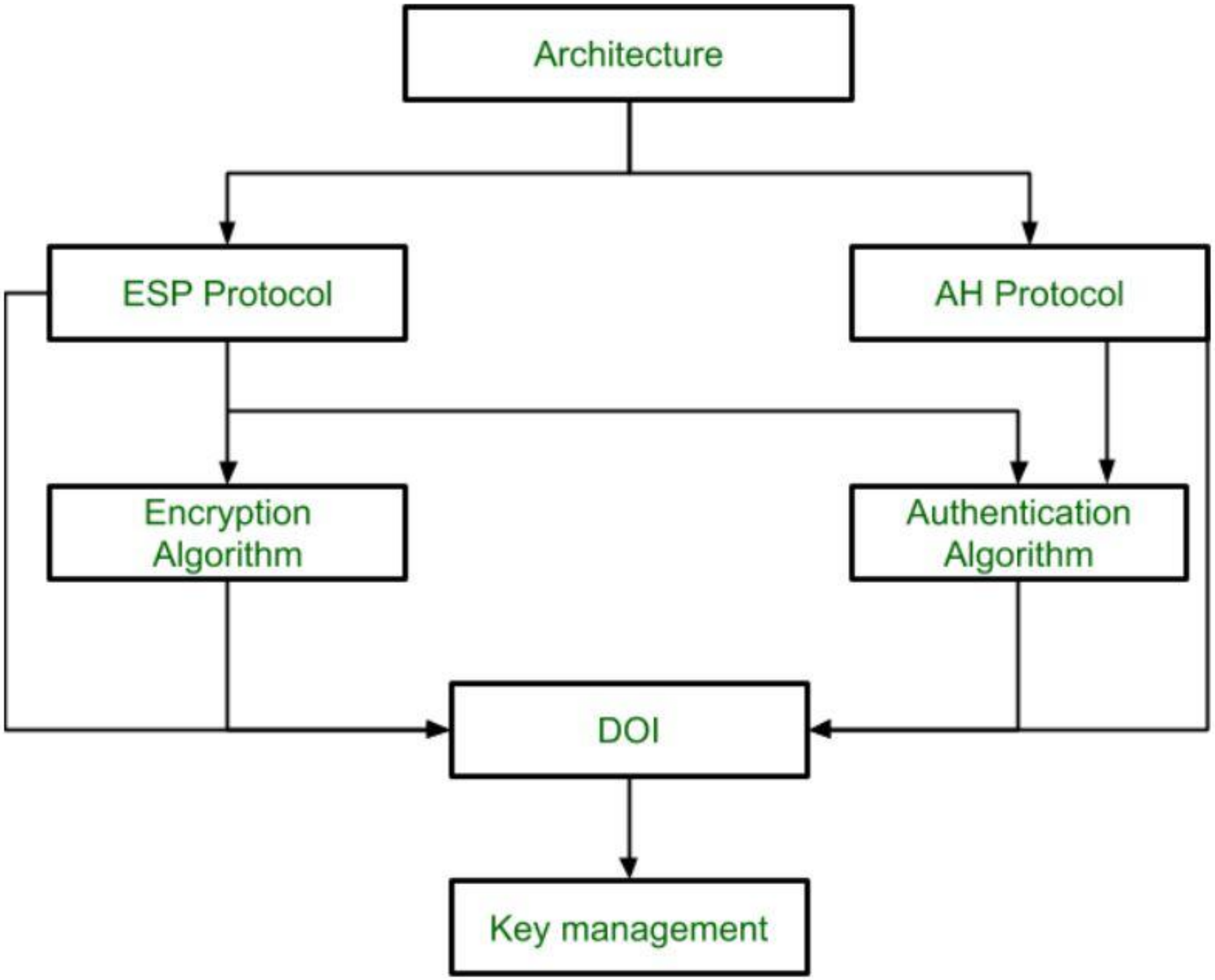
IPSec (IP Security) Architecture

IPSec (IP Security) Architecture

IPSec (IP Security) architecture uses two protocols to secure the traffic or data flow. These protocols are ESP (Encapsulation Security Payload) and AH (Authentication Header). IPSec Architecture includes protocols, algorithms, DOI, and Key Management. All these components are very important in order to provide the three main services:

- Confidentiality
- Authentication
- Integrity

IPSec (IP Security) Architecture



IPSec (IP Security) Architecture

1. **Architecture:** Architecture or IP Security Architecture covers the general concepts, definitions, protocols, algorithms, and security requirements of IP Security technology.

2. **ESP Protocol:** ESP (Encapsulation Security Payload) provides a confidentiality service.

3. **Encryption algorithm:** The encryption algorithm is the document that describes various encryption algorithms used for Encapsulation Security Payload.

4. **AH Protocol:** AH (Authentication Header) Protocol provides both Authentication and Integrity services.

IPSec (IP Security) Architecture

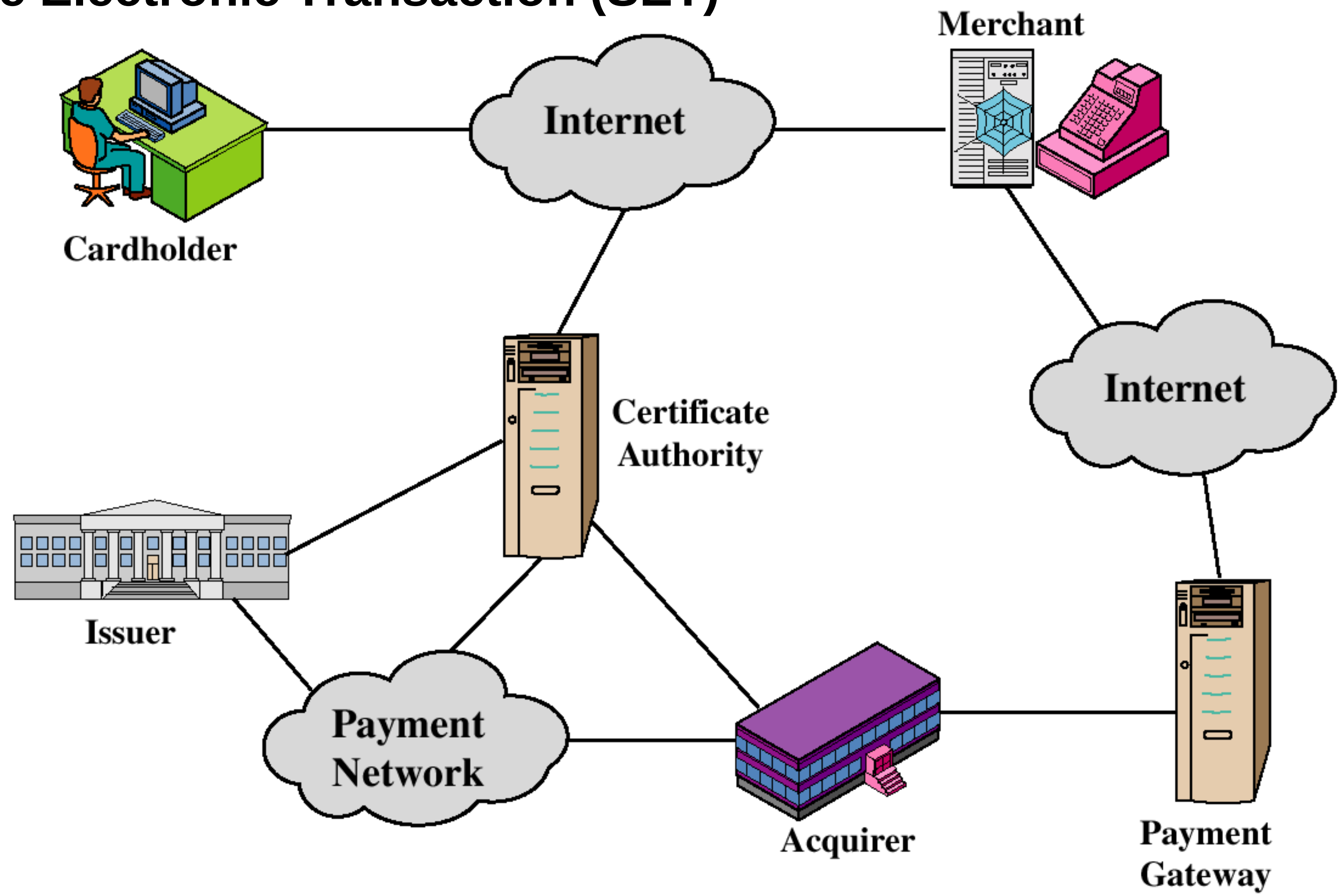
5. **Authentication Algorithm:** The authentication Algorithm contains the set of documents that describe the authentication algorithm used for AH and for the authentication option of ESP.

6. **DOI (Domain of Interpretation):** DOI is the identifier that supports both AH and ESP protocols. It contains values needed for documentation related to each other.

7. **Key Management:** Key Management contains the document that describes how the keys are exchanged between sender and receiver.

Handshaking

Secure Electronic Transaction (SET)



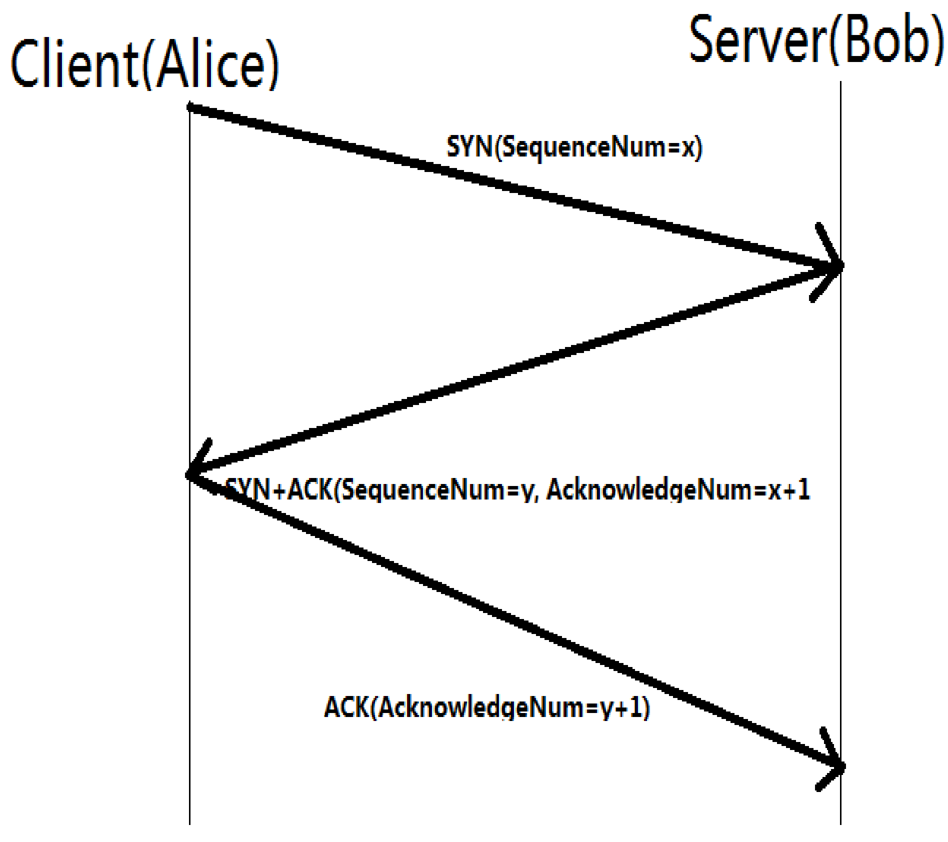
Handshaking

Handshaking

Handshaking is an automated process of negotiation that dynamically sets parameters of a communications channel established between two entities before normal communication over the channel begins. It follows the physical establishment of the channel and precedes normal information transfer.

Handshaking

Handshaking




Thank You!